Demo Client · DEMO-0008
11 November 2026 · v1.0

| Field | Value |
|---|---|
| Document | Methodology of the transaction cost analysis |
| Template version | transaction_costs 1.0.0 / proquant-core 0.1.0 |
| Engagement | DEMO-0008 · Demo Client |
| Period | 2024-04-01 to 2024-06-14 |
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## 1. Purpose

The analysis measures what it cost to turn the portfolio managers' decisions into trades, where the cost came from, who executed better or worse once the difficulty of the orders is taken into account, whether the best-execution policy was respected, and what could be saved. It is a measurement of execution, not of the quality of the investment decisions.

## 2. Inputs

| Data set | Size | Fingerprint (SHA-256) |
|---|---|---|
| orders | 1,171 rows | `385479b70739727c` |
| fills | 12,737 rows | `9f474f09f7fed8bf` |
| bars | 85,566 rows | `944f1f24550488b9` |

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail.

## 3. Cleaning

Side is read from B / Buy / S / Sell spellings; symbols are upper-cased and an exchange suffix removed; timestamps are parsed. Exact duplicate fills are removed and counted. An order is **excluded** from the cost analysis, and listed with its reason, when it cannot be measured honestly: unknown side, time or quantity; arrival before decision; no fills; filled for more than its quantity; no market data around its execution; a fill earlier than the arrival (a clock problem); a fill outside market hours; a fill priced outside the high-low range of its bar by more than a tolerance. A missing decision price does not exclude an order: the arrival price is used and the order is flagged.

## 4. Benchmarks

The price at a time is the open of the 5-minute bar that contains it. For each order: the **decision price** D (as recorded), the **arrival price** A (the bar of the arrival time), the **interval VWAP** (typical price (high + low + close) / 3 weighted by volume, from the bar of the arrival to the bar of the last fill), the **close** C of the day and the **post-trade price** (30 minutes after the last fill). Average daily volume and daily volatility use the previous 20 trading days only.

## 5. Implementation shortfall

With side s (+1 buy, -1 sell), order quantity Q, executed quantity q = sum q_i at prices p_i and the paper notional Q x D:

- **delay** = s x q x (A - D): the market moved while the order waited;
- **execution** = s x sum q_i (p_i - A): what the desk and the broker paid against the arrival price;
- **opportunity** = s x (Q - q) x (C - D): the unfilled part, valued at the close;
- **explicit** = commissions + fees.

The shortfall is their sum in basis points of the paper notional; the four parts add up exactly. Other measures: slippage against the arrival price and against the interval VWAP per share executed; **reversion** = s x (average price - post-trade price) / average price (positive: the price moved back, a sign of temporary impact); **participation** = executed quantity / market volume of the execution interval; order size as a share of ADV. Positive numbers are costs. Averages are weighted by the paper notional.

## 6. The pre-trade model and the fair comparison

Slippage against arrival is explained by sigma x sqrt(size / ADV) (the square-root law of market impact, sigma = daily volatility in bps), half the quoted spread and urgency dummies, by ordinary least squares on the first 70% of the trading days (the remaining days test it); the extreme 1% of costs on each side is clipped before the fit. The **expected cost** of an order is the fitted value; its **alpha** is the actual slippage minus the expected cost (positive: worse than expected for an order of that difficulty). The R-squared of cost models on single orders is low because the market moves during execution by much more than the impact of the order; the model is judged on the sign and significance of its size coefficient and used to compare groups, not to quote the cost of one order.

## 7. Groups and confidence

Costs are summarised by broker, algorithm, broker and algorithm, order size, time of arrival, desk, urgency, side, month, spread and order type, and by venue at fill level. For each group the alpha has a 95% **bootstrap** interval (resampling orders, weighted by notional) and p-value; a group is "worse (better) than the model" only if the interval excludes zero, "not significant" otherwise, and "too few orders" below five orders.

## 8. Best-execution policy

Limits agreed at kickoff and recorded in the configuration: no fill beyond the limit price (tolerance in bps), slippage against arrival by urgency, participation, time from decision to arrival, commissions in bps, share of notional with one broker, decision price recorded. Every breach is an exception with the order, the value and the limit; exceptions go to the order review sheet.

## 9. Savings

Scenarios, annualised by the number of trading days: brokers and algorithms above the median alpha brought to the median; orders above a cap on the share of ADV split (saving = model impact coefficient x sigma x (sqrt(size) - sqrt(cap))); slower desks brought to the delay cost of the fastest; commissions above the cap brought to the cap; part of the opening half hour moved to midday. The scenarios overlap: the sum is an upper bound and a realistic total applies an overlap factor.

## 10. Assessment

Twenty-two tests in five areas (data, cost, brokers, policy, impact) give PASS, WARN or FAIL against thresholds agreed at kickoff; the overall view is Good (all pass), Acceptable with actions (some warn) or Material concerns (some fail), with the action each non-passing test implies. A broker's alpha counts only when it is statistically significant.

## 11. Commentary

Sentences are assembled by rules from the tables: headline, benchmarks and reversion, brokers, algorithms, size and the model, timing and desks, policy, data, savings and the overall view. Where a number is missing the sentence says so. The analyst reads the draft and may add notes.

## 12. Reproducibility

Inputs, configuration and code version are recorded in `audit.json`. The bootstrap uses a fixed seed. Every table in the deliverables is a CSV in the run folder.

## 13. Limitations

The limits of this work are set out in the report. No deviations from the standard methodology.